

4.2.8. Titanic Dataset Analysis and Data Cleaning - 401:49

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Get the number of survivors by gender (Sex).
2. Get the number of non-survivors by gender (Sex).
3. Get the number of survivors by embarkation location (Embarked_S).
4. Get the number of non-survivors by embarkation location (Embarked_S).
5. Calculate the percentage of children (Age < 18) who survived.
6. Calculate the percentage of adults (Age >= 18) who survived.
7. Get the median age of survivors.
8. Get the median age of non-survivors.
9. Get the median fare of survivors.
10. Get the median fare of non-survivors.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	ParCh	Ticket	Fare	Cabin	Embarked

Sample Data:

titanicDat...SUBMIT

1import pandas as pd
2import numpy as np
3
4# Load the Titanic dataset
5data = pd.read_csv('Titanic-Dataset.csv')
6data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
7
8
9# 1. Get the number of survivors by gender
10print(data[data['Survived']==1]['Sex'].value_counts())
11
12
13# 2. Get the number of non-survivors by gender
14print(data[data['Survived']==0]['Sex'].value_counts())
15
16# 3. Get the number of survivors by embarked location
17print(data[data['Survived']==1]['Embarked_S'].value_counts())
18
19# 4. Get the number of non-survivors by embarked location
20print(data[data['Survived']==0]['Embarked_S'].value_counts())
21
22# 5. Calculate the percentage of children (Age < 18) who survived
23print(data[data['Age']<18]['Survived'].mean())
24

TerminalTest cases

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5.1.1. Stacked Plot18:49

Create a stacked area plot to visualize the temperature variations for three different cities (City A, City B, and City C) across the months of the year. The temperature data is provided for each city in the editor.

Your task is to:

- Create a stacked area plot using the data.
- Label the x-axis as "Month", the y-axis as "Temperature", and provide the title "Temperature Variation" for the plot.
- Display the plot showing the temperature variation for each city throughout the months of the year.

Sample Test Cases

stackedpl...

```
1 import matplotlib.pyplot as plt
2 import pandas as pd
3
4 # Data for Months and Temperature for three cities
5 data = {
6     'Month': ['January', 'February', 'March', 'April', 'May', 'June',
7             'July', 'August', 'September', 'October', 'November', 'December'],
8     'City_A_Temperature': [5, 7, 10, 13, 17, 20, 22, 21, 18, 12, 8, 6],
9     'City_B_Temperature': [2, 3, 5, 6, 10, 14, 16, 17, 12, 9, 5, 3],
10    'City_C_Temperature': [3, 4, 6, 8, 9, 12, 15, 14, 10, 7, 4, 2]
11 }
12
13 plt.stackplot(data['Month'], data['City_A_Temperature'], data['City_B_Temperature'], data['City_C_Temperature'])
14 plt.xlabel('Month')
15 plt.ylabel('Temperature')
16 plt.title('Temperature Variation')
17 plt.show()
```

Terminal Test cases

3 32°C Sunny

Search

ENG IN 61% 11:54 AM 06-05-2026

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5.2.1. Titanic Dataset

01:24

Write a Python program to analyze and visualize data from the Titanic dataset based on the following instructions:

Dataset Information:

The dataset is stored in a CSV file named `titanic.csv` and has been loaded using the `pandas` library. It contains the following columns:

- `Pclass`: Passenger class (1 = First, 2 = Second, 3 = Third).
- `Gender`: Gender of the passenger (male/female).
- `Age`: Age of the passenger.
- `Survived`: Survival status (0 = Did not survive, 1 = Survived).
- `Fare`: Ticket fare paid by the passenger.

Visualization:

To represent these trends, you will create 5 visualizations using `Matplotlib`. The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns).

Visualization Details:

Write the code to create a series of visualizations as follows:

Bar Plot (Pclass Distribution):

- Create a bar plot to show the distribution of passengers across the different passenger classes (`Pclass`).

Explorer

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```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset from the CSV file
df = pd.read_csv('titanic.csv')

# Set up the figure for 5 subplots
fig, axes = plt.subplots(3, 2, figsize=(12, 12))

# write the code..
count = df['Pclass'].value_counts().sort_index()
count_gen = df['Gender'].value_counts()
count_s = df['Survived'].value_counts().sort_index()

axes[0,0].bar(count.index, count.values, color='skyblue')
axes[0,0].set_title('Passenger Class Distribution')
axes[0,0].set_xlabel('Pclass')
axes[0,0].set_ylabel('Count')

axes[0,1].pie(count_gen.values, labels=['male', 'female'], colors=
['lightblue', 'lightcoral'], autopct='%1.1f%%')
axes[0,1].set_title('Gender Distribution')
```

Terminal

Test cases

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NEXT

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5.2.2. Histogram of passenger information of Titanic

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

- Use 30 bins for the histogram.
- Set the edge color of the bars to black (k).
- Label the x-axis as 'Age' and the y-axis as 'Frequency'.
- Add the title "Age Distribution" to the histogram.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17590,71.2832,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,573458,8.00,,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4503,,Q
```

Histogram...

1 import pandas as pd

2 import matplotlib.pyplot as plt

3

4 # Load the Titanic dataset

5 data = pd.read_csv('Titanic-Dataset.csv')

6

7 # Data Cleaning

8 data['Age'].fillna(data['Age'].median(), inplace=True)

9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

10 data.drop('Cabin',axis=1, inplace=True)

11

12 # Convert categorical features to numeric

13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

15

16 # Write your code here for Histogram

17 plt.hist(data['Age'],bins=30,edgecolor='k')

18 plt.title('Age Distribution')

19 plt.xlabel('Age')

20 plt.ylabel('Frequency')

21 plt.show()

Terminal

Test cases

5.2.3. Bar plot of survival rate of passengers

Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:

1. Use the 'Survived' column to show the count of survivors (0 = Did not survive, 1 = Survived).
2. Set the chart type to 'bar'.
3. Add the title "Survival Count" to the chart.
4. Label the x-axis as 'Survived' and the y-axis as 'Count'.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17590,71.2832,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373458,8.05,,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,,Q
```

BarPlotOf...

1

import pandas as pd

2

import matplotlib.pyplot as plt

3

4

Load the Titanic dataset

5

data = pd.read_csv('Titanic-Dataset.csv')

6

7

Data Cleaning

8

data['Age'].fillna(data['Age'].median(), inplace=True)

9

data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

10

data.drop('Cabin', axis=1, inplace=True)

11

12

Convert categorical features to numeric

13

data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

14

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

15

16

Write your code here for Bar Plot for Survival Rate

17

survival_counts = data['Survived'].value_counts()

18

survival_counts.plot(kind='bar')

19

plt.title('Survival Count')

20

plt.xlabel('Survived')

21

plt.ylabel('Count')

22

23

plt.show()

24

Terminal

Test cases

5.2.4. Bar Plot for Survival by Gender

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the 'Sex' column, then use the value_counts() function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.

2. Use a stacked bar chart to display the survival counts.

3. Add the title "Survival by Gender" to the chart.

4. Label the x-axis as 'Gender' and the y-axis as 'Count'.

5. The legend should indicate 'Not Survived' and 'Survived'.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,42,1,0,A/5 21171,7.25,,S
2,1,1,"Cummings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2, 3101282,7.92,,S

BarPlotOf...

1import pandas as pd
2import matplotlib.pyplot as plt
3
4# Load the Titanic dataset
5data = pd.read_csv('Titanic-Dataset.csv')
6
7# Data Cleaning
8data['Age'].fillna(data['Age'].median(), inplace=True)
9data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10data.drop('Cabin', axis=1, inplace=True)
11
12# Convert categorical features to numeric
13data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16# Write your code here for Bar Plot for Survival by Gender
17survival_counts = data.groupby('Sex')['Survived'].value_counts().unstack()
18survival_counts.plot(kind='bar', stacked=True)
19
20plt.title('Survival by Gender')
21plt.xlabel('Gender')
22plt.ylabel('Count')
23plt.legend(['Not Survived', 'Survived'])
24plt.show()

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5.2.5. Bar Plot for Survival by Pclass

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (Pclass), in the Titanic dataset. The chart should display the following specifications:

- Group the data by the Pclass column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using value_counts().
- Use a stacked bar chart to display the survival counts.
- Add the title "Survival by Pclass" to the chart.
- Label the x-axis as 'Pclass' and the y-axis as 'Count'.
- The legend should indicate 'Not Survived' and 'Survived'.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,42,1,0,A/5 21171,7.25,,S
2,1,1,"Cummings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2, 3101282,7.92,,S
```

BarPlotOf...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Bar Plot for Survival by Pclass
17 survival_counts = data.groupby('Pclass')
18 ['Survived'].value_counts().unstack()
19 survival_counts.plot(kind='bar', stacked=True)
20
21 plt.title('Survival by Pclass')
22 plt.xlabel('Pclass')
23 plt.ylabel('Count')
24 plt.legend(['Not Survived', 'Survived'])
```

Terminal Test cases

5.2.6. Bar Plot for Survival by Embarked12:29

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset.
The chart should display the following specifications:
1. Use the **Embarked** column to determine the embarkation location. After converting this column into dummy variables (using `pd.get_dummies()`), plot the survival count based on the **Embarked_Q** column (representing passengers who embarked from Queenstown) in relation to survival.
2. Set the chart type to 'bar' and make it stacked.
3. Add the title "Survival by Embarked " to the chart.
4. Label the x-axis as 'Embarked' and the y-axis as 'Count'.
5. Include a legend to distinguish between survivors and non-survivors (label the legend as 'Survived' and 'Not Survived').

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	ParCh	Ticket	Fare	Cabin	Embarked

Sample Data:

BarPlotOf...SUBMIT

1import pandas as pd
2import matplotlib.pyplot as plt
3
4# Load the Titanic dataset
5data = pd.read_csv('Titanic-Dataset.csv')
6
7# Data Cleaning
8data['Age'].fillna(data['Age'].median(), inplace=True)
9data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10data.drop('Cabin', axis=1, inplace=True)
11
12# Convert categorical features to numeric
13data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16# Write your code here for Bar Plot for Survival by Embarked
17sc = data.groupby('Embarked_Q')['Survived'].value_counts().unstack()
18sc.plot(kind='bar', stacked=True)
19plt.title('Survival by Embarked')
20plt.xlabel('Embarked')
21plt.ylabel('Count')
22plt.legend(['Not Survived', 'Survived'])
23plt.show()

TerminalTest cases

5.2.7. Box plot for Age Distribution

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset across different passenger classes. The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Age by Pclass"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17590,71.2832,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373458,8.05,,S
6,0,3,"Moran, Mr. James",male,0,0,330877,0.4586,,Q
```

BoxPlotF...

1

import pandas as pd

2

import matplotlib.pyplot as plt

3

4

Load the Titanic dataset

5

data = pd.read_csv('Titanic-Dataset.csv')

6

7

Data Cleaning

8

data['Age'].fillna(data['Age'].median(), inplace=True)

9

data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

10

data.drop('Cabin', axis=1, inplace=True)

11

12

Convert categorical features to numeric

13

data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

14

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

15

16

Write your code here for Box Plot for Age by Pclass

17

18

data.boxplot(column='Age', by='Pclass')

19

plt.title('Age by Pclass')

20

plt.suptitle('')

21

plt.xlabel('Pclass')

22

plt.ylabel('Age')

23

plt.show()

24

Terminal

Test cases

5.2.8. Box Plot for Age by Survived04:11

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:

1. Use the **Survived** column to group the data for the boxplot (0 = Did not survive, 1 = Survived).
2. Set the title of the plot to "Age by Survival".
3. Remove the default subtitle with `plt.suptitle("")`.
4. Label the x-axis as 'Survived' and the y-axis as 'Age'.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cummings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Petrucci, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,873458,8.05,,S

BoxPlotF...SUBMIT

1import pandas as pd
2import matplotlib.pyplot as plt
3
4# Load the Titanic dataset
5data = pd.read_csv('Titanic-Dataset.csv')
6
7# Data Cleaning
8data['Age'].fillna(data['Age'].median(), inplace=True)
9data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10data.drop('Cabin', axis=1, inplace=True)
11
12# Convert categorical features to numeric
13data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16# Write your code here for Box Plot for Age by Survived
17data.boxplot(column='Age', by='Survived')
18plt.title('Age by Survival')
19plt.suptitle('')
20plt.xlabel('Survived')
21plt.ylabel('Age')
22plt.show()

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5.2.9. Box Plot for Fare by Pclass

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:

- Use the Pclass column to group the data for the boxplot.
- Set the title of the plot to "Fare by Pclass".
- Remove the default subtitle with plt.suptitle('').
- Label the x-axis as 'Pclass' and the y-axis as 'Fare'.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17590,71.2832,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373458,8.05,,S
6,0,3,"Moran, Mr. James",male,0,0,330877,0.4586,,Q
```

BoxPlotF...

1

import pandas as pd

2

import matplotlib.pyplot as plt

3

4

Load the Titanic dataset

5

data = pd.read_csv('Titanic-Dataset.csv')

6

7

Data Cleaning

8

data['Age'].fillna(data['Age'].median(), inplace=True)

9

data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

10

data.drop('Cabin', axis=1, inplace=True)

11

12

Convert categorical features to numeric

13

data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

14

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

15

16

Write your code here for Box Plot for Fare by Pclass

17

data.boxplot(column='Fare', by='Pclass')

18

plt.title('Fare by Pclass')

19

plt.suptitle('')

20

plt.xlabel('Pclass')

21

plt.ylabel('Fare')

22

plt.show()

23

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Test cases

5.2.10. Scatter Plot for Age vs. Fare

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Set the title of the plot to **"Age vs. Fare"**.
3. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 11175,7.25,,S
2,1,1,"Cummings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17509,53.00,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.0,,S
6,0,3,"Monaghan, Mr. James",male,0,0,538077,0.0,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.0,,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,1,1,344004,31.0,,S
```



5.2.11. Scatter Plot for Age vs. Fare by Survived

01:09

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.

2. Color the points based on the **Survived** column: **Red** for passengers who did not survive (**Survived = 0**). **Blue** for passengers who survived (**Survived = 1**).

3. Set the title of the plot to **"Age vs. Fare by Survival"**.

4. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cummings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.3833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

AgeFareS...

SUBMIT

1 import pandas as pd

2 import matplotlib.pyplot as plt

3

4 # Load the Titanic dataset

5 data = pd.read_csv('Titanic-Dataset.csv')

6

7 # Data Cleaning

8 data['Age'].fillna(data['Age'].median(), inplace=True)

9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

10 data.drop('Cabin', axis=1, inplace=True)

11

12 # Convert categorical features to numeric

13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

15

16 # Write your code here for Scatter Plot for Age vs. Fare by Survived

17

18 colors = data['Survived'].map({0: 'red', 1: 'blue'})

19 plt.scatter(data['Age'], data['Fare'], c=colors)

20 plt.title('Age vs. Fare by Survival')

21 plt.xlabel('Age')

22 plt.ylabel('Fare')

23 plt.show()

24

Terminal Test cases